Mouadh Bondka

Software Engineer | Full Stack and Applied AI

contact@moudka.com

portfolio.moudka.com | github.com/Cordoncharge83

IMT Atlantique graduate with experience delivering Java/Spring software for CNAM healthcare systems and building a multimodal RAG assistant at La Banque Postale. Independently builds and deploys web and desktop products, including VALORANT analytics, an AI negotiation coach, a searchable video archive, and tournament software. Works across backend services, React/TypeScript interfaces, data pipelines, testing, and deployment, with additional Unity/C# gameplay experience.

Professional experience

Sopra Steria

Software Engineering Intern | April 2025 to October 2025

- Delivered Java/Spring Boot features across production releases for ARPEGE, part of CNAM's healthcare information systems serving millions of users.
- Investigated production incidents across business rules, SQL queries, application logs, and a Java codebase exceeding 200,000 lines; implemented and tested corrective changes.
- Developed a Spring Batch prototype for bulk case re-entry, designed for up to 9,000 records, with CSV input, service orchestration, data mapping, error handling, counters, and execution reports.
- Validated changes with JUnit, Mockito, and SoapUI service mocks, maintaining over 80% automated test coverage on critical components and working within SonarQube quality gates.
- Collaborated with functional analysts, testers, and developers on investigation, implementation, validation, and delivery in an enterprise Java and AngularJS environment.

La Banque Postale

AI Engineering Intern | May 2024 to August 2024

- Built a multimodal RAG desktop assistant for searching internal documentation, covering document ingestion, retrieval, and a custom user interface.
- Implemented document processing, embeddings, PostgreSQL/pgvector storage, and retrieval pipelines with Python and LangChain, supporting text, tables, and images.
- Evaluated retrieval and re-ranking strategies for relevance and latency, then delivered the prototype with documentation and a technical handoff.

Education

IMT Atlantique

Engineering Degree in Software Engineering | Nantes, France | September 2022 to October 2025

Master's level engineering degree. Focus on software architecture, distributed systems, microservices, DevOps, functional programming, and advanced algorithms.

IPEST

Preparatory Classes for Engineering Schools | Tunis, Tunisia | 2020 to 2022

Intensive preparatory studies in mathematics, physics, and engineering sciences.

Independent software projects

Comp Atlas

VALORANT analytics and opponent preparation | Working project name | 2026

- Built and deployed a VALORANT research workspace combining composition discovery, team scouting, match records, and source evidence for coaches and analysts.
- Implemented exact lineup and agent-core searches, composition rankings, non-mirror win rates, team map pools, veto analysis, and opponent comparisons.
- Added round and economy analysis, including pistol conversions, second-round recovery, bonus outcomes, and force-buy statistics, with coverage and sample-size indicators.
- Developed a TypeScript import workflow with batched collection, pause/resume, validation, and review before publication; immutable D1/R2 snapshots support revision history and restoration.
- Preserved research context through shared filters and URLs, and used snapshot caching and ETags to reduce repeated data processing on Cloudflare Workers.

React, TypeScript, Cloudflare Workers, D1, R2

Smosh Story Finder

Searchable YouTube story archive | 2026

- Built and deployed a React/TypeScript archive for finding individual Smosh Reddit stories by guest, title, and subreddit, with timestamped links to the original videos.
- Implemented shareable filter state, episode grouping, incremental rendering, and a responsive guest carousel with mouse, touch, and keyboard controls.
- Created a collect-review-publish workflow using YouTube metadata, a standalone review tool, reviewed ZIP imports, and GitHub Actions, with duplicate prevention and an import ledger.
- Tested filters, URL restoration, gestures, keyboard focus, and incremental results; deployed on Cloudflare Pages with contextual link previews.

React, TypeScript, Vite, GitHub Actions, Cloudflare Pages

smosh-story-finder.moudka.com

DuelSync

Offline tournament desktop companion and public results

- Built a Yu-Gi-Oh! tournament companion with a Tauri/React desktop interface, packaged FastAPI backend, and SQLite persistence.
- Implemented persistent Konami Tournament Software file monitoring, round and pairing synchronization, standings, BYE handling, and configurable Top Cut brackets with standings-based seeding.
- Added backup and restoration workflows, keeping core tournament operations available offline and synchronizing public results when connectivity returns.
- Published live results through Cloudflare Workers, KV, and Pages with QR-code sharing and player-data protection; iterated with a tournament organizer.

Tauri, React, Python, FastAPI, SQLite, Cloudflare Workers, KV, Pages

duelsync.moudka.com

Bargainy

AI negotiation coach

- Designed and deployed multi-turn job-offer, rent, and purchase simulations with personalized feedback, a Quick Coach mode, and an interactive Spline assistant.
- Separated private user objectives from the simulated counterpart's context; integrated server-side Gemini requests with structured outputs, model fallbacks, protected credentials, and error handling.

React, TypeScript, Gemini API, Spline | bargainy.moudka.com

Game development and additional projects

CapsuleMonstersGX

Tactical RPG prototype | Unity and C#

- Designed and implemented grid-based tactical combat with finite-state-machine battle flow, movement and attack validation, summoning, action-point management, fusion, and turn progression.
- Built enemy decision-making for summon selection, target prioritization, tactical movement, and attack execution, evaluating kill potential, expected damage, distance, and threat.
- Created reusable unit, statistics, selection, interaction, and combat-feedback systems, connected through events to gameplay UI updates.

VR Renewable Energy Simulation

Academic team project | Unity and C#

- Contributed gameplay programming within a multidisciplinary team of five to six students building an educational virtual reality experience.
- Implemented player movement, object interactions, level progression, and NPC dialogue, including dialogue options unlocked by player actions and completed objectives.

Game Recommender

Semantic recommendation web application | Python, FastAPI, React, PostgreSQL and pgvector

- Built embedding generation and vector-similarity retrieval pipelines, exposing game recommendations through a FastAPI REST API and React interface.

Leadership

WAKE at IMT Atlantique

President and Head of Public Relations | February 2023 to November 2023

- Led a 20-member organizing committee delivering a student event for up to 500 participants; coordinated partnerships, communications, sponsorships, logistics, and stakeholder relationships.

Technical skills

Programming: Java, Python, TypeScript, JavaScript, C#, SQL; additional experience with C++, C, and Haskell.

Backend and data: Spring Boot, Spring Batch, FastAPI, REST APIs, OpenAPI, PostgreSQL, pgvector, SQLite.

Frontend and desktop: React, AngularJS, Tauri, HTML/CSS, Vite.

AI and retrieval: LangChain, RAG, embeddings, semantic search, vector databases, multimodal retrieval, re-ranking, LLM integration.

Testing and delivery: JUnit, Mockito, SoapUI, SonarQube, Git, GitHub Actions, Docker, Cloudflare Workers, Pages, D1, R2, KV.

Gameplay: Unity, tactical AI, state machines, event-driven systems, gameplay UI, VR interactions.

Technical writing

Building Smosh Story Finder: The Engineering Behind a Searchable YouTube Archive — published on Hashnode, October 2026.

smosh-stories-finder-tech.hashnode.dev

Languages

Arabic — Native | French — C1 | English — C1 | German — A2 | Spanish — A2